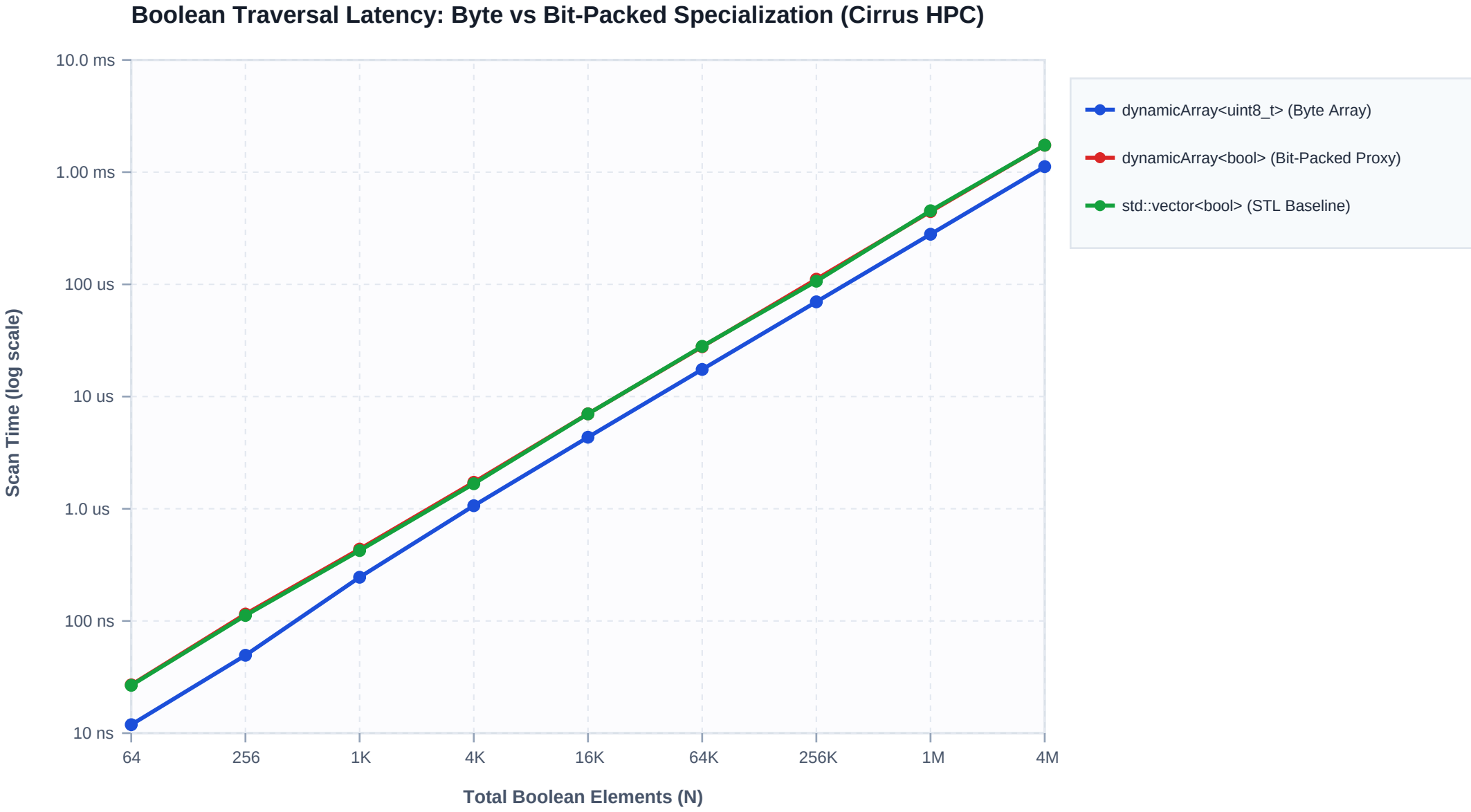
Boolean Traversal Latency: Byte vs Bit-Packed Specialization (Cirrus HPC)
Scan Time (log scale)
10.0 ms
1.00 ms
100 us
10 us
1.0 us
100 ns
10 ns
64
256
1K
4K
16K
64K
256K
1M
4M
Total Boolean Elements (N)
dynamicArray<uint8_t> (Byte Array)
dynamicArray<bool> (Bit-Packed Proxy)
std::vector<bool> (STL Baseline)